

Lecture-02

0612-302 Data Communication

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Chapter-2

Network Models

Textbook: Data Communications and Networking
Fourth Edition
- Behrouz A. Forouzan

Introduction

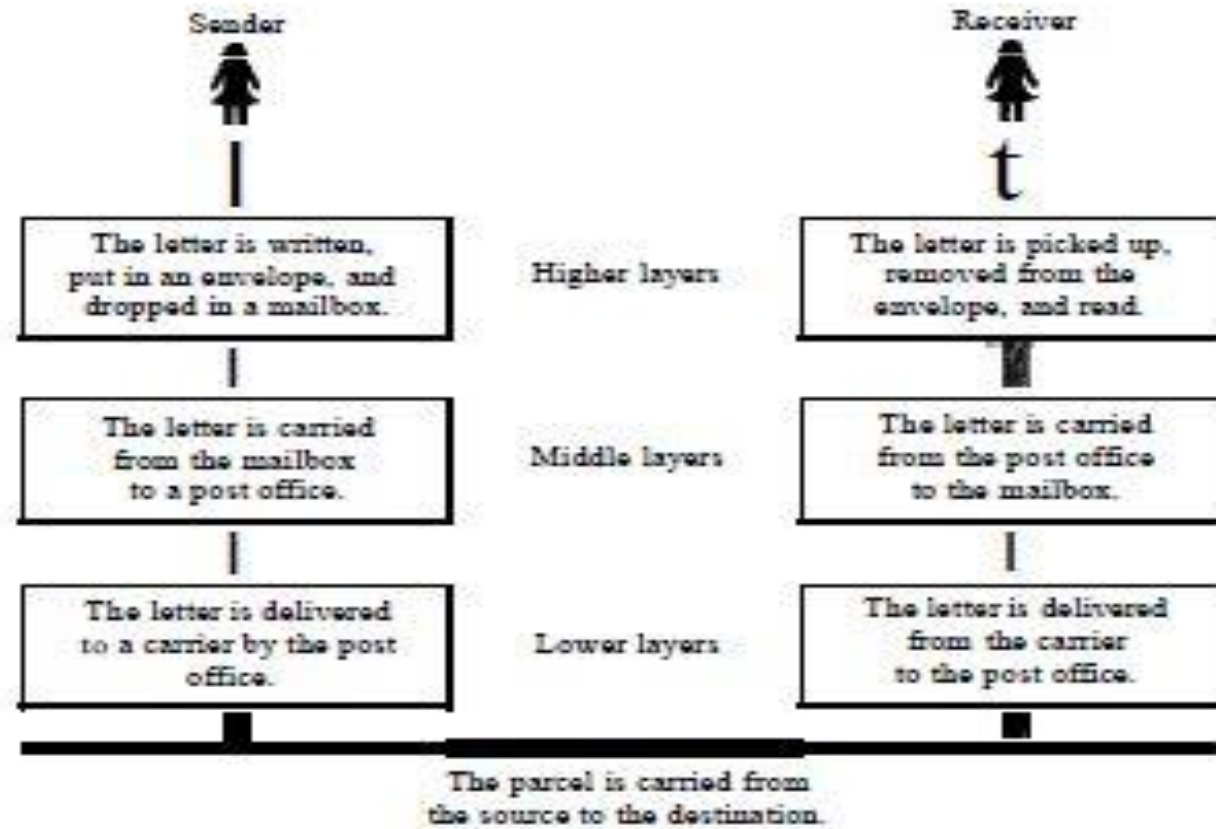
- A network is a combination of hardware and software that sends data from one location to another.
- The hardware consists of the physical equipment that carries signals from one point of the network to another.
- The software consists of instruction sets that make possible the services that we expect from a network.

Layered Tasks

- We use the concept of layers in our daily life.
- As an example, let us consider two friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.
- Figure 2.1 shows the steps in this task.

Layered Tasks

Figure 2.1 *Tasks involved in sending a letter*



Layered Tasks

- **Sender, Receiver, and Carrier**
- In Figure 2.1 we have a sender, a receiver, and a carrier that transports the letter. There is a hierarchy of tasks.

At the Sender Site:

- Let us first describe, in order, the activities that take place at the sender site.
 - Higher layer: The sender writes the letter, inserts the letter in an envelope, writes the sender and receiver addresses, and drops the letter in a mailbox.
 - Middle layer: The letter is picked up by a letter carrier and delivered to the post office.
 - Lower layer: The letter is sorted at the post office; a carrier transports the letter.

Layered Tasks

On the Way

- The letter is then on its way to the recipient.
- On the way to the recipient's local post office, the letter may actually go through a central office.
- In addition, it may be transported by truck, train, airplane, boat, or a combination of these.

At the Receiver Site

- Lower layer: The carrier transports the letter to the post office.
- Middle layer: The letter is sorted and delivered to the recipient's mailbox.
- Higher layer; The receiver picks up the letter, opens the envelope, and reads it.

The OSI Model

- Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards.
- An ISO standard that covers all aspects of network communications is the Open Systems Interconnection model.
- It was first introduced in the late 1970s. An open system is a set of protocols that allows any two different systems to communicate regardless of their underlying architecture.
- The purpose of the OSI model is to show how to facilitate communication between different systems without requiring changes to the logic of the underlying hardware and software.

The OSI Model

- The OSI model is not a protocol; it is a model for understanding and designing a network architecture that is flexible, robust, and interoperable.
- The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.
- It consists of seven separate but related layers, each of which defines a part of the process of moving information across a network.

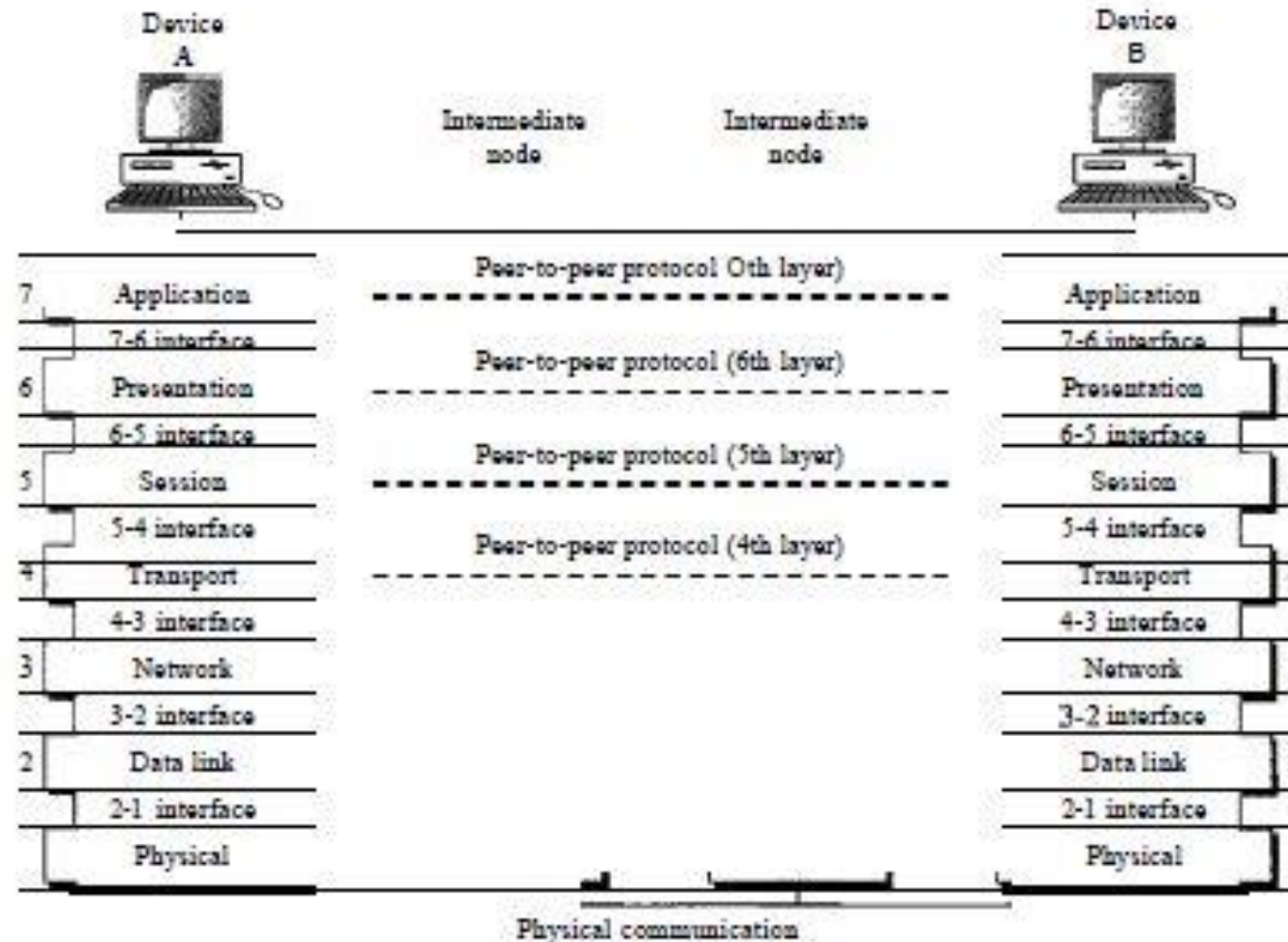
The OSI Model

7 Layers of OSI Model

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data link Layer
7. Physical Layer

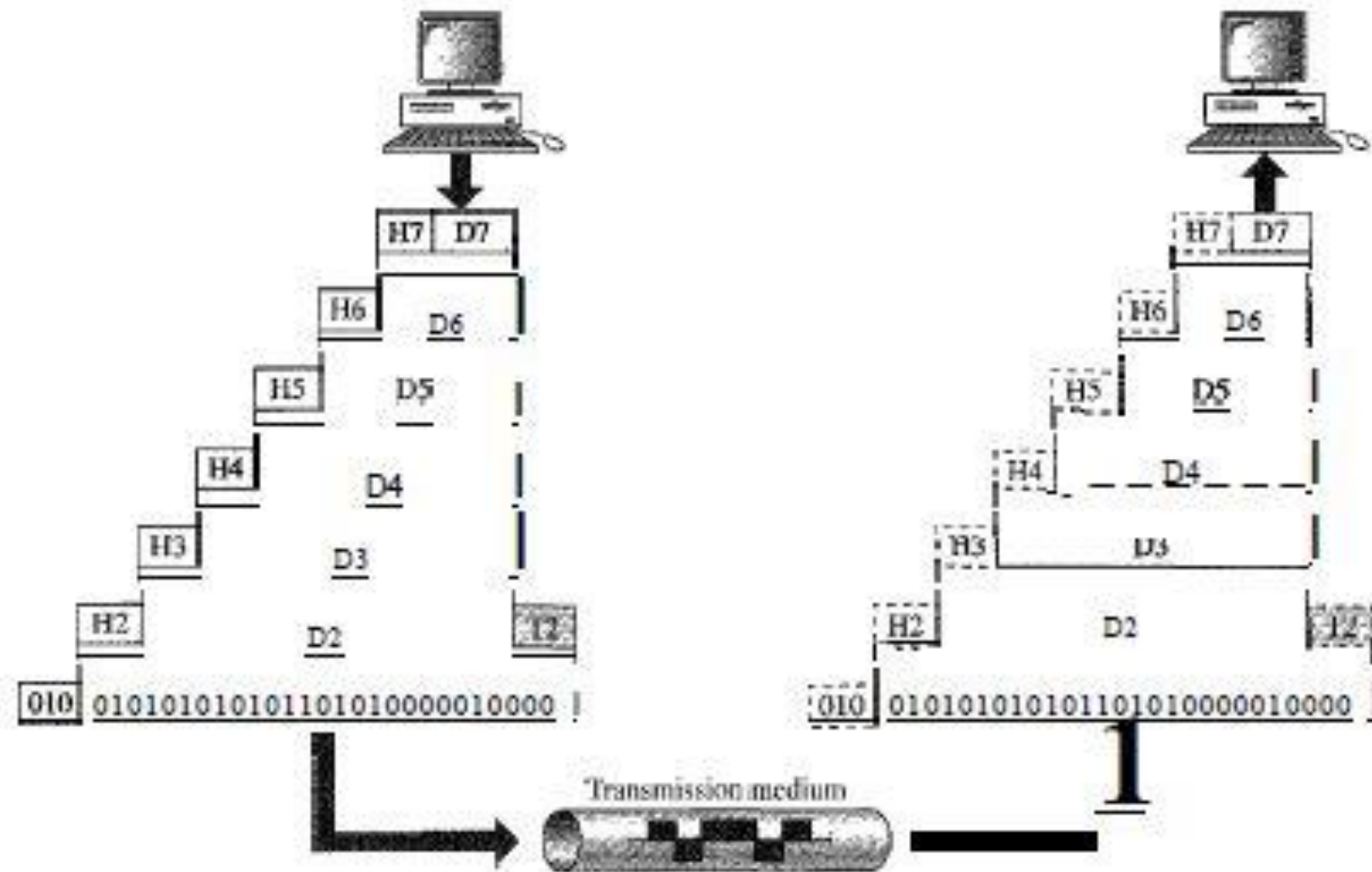
The OSI Model

Figure 2.3 *The interaction between layers in the OSI model*



The OSI Model

Figure 2.4 *An exchange using the OSI model*



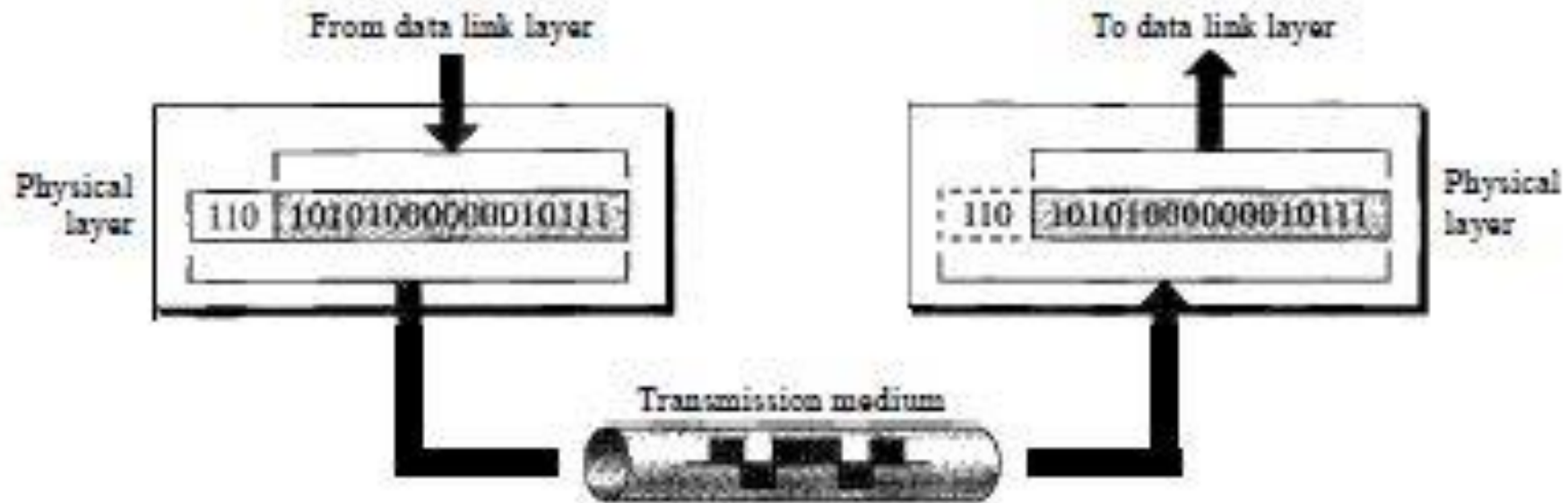
Physical Layer

➤ Physical Layer

- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- It deals with the mechanical and electrical specifications of the interface and transmission medium.
- It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to Occur.
- Figure 2.5 shows the position of the physical layer with respect to the transmission medium and the data link layer.

Physical Layer

Figure 2.5 *Physical layer*



The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Physical Layer

The physical layer is also concerned with the following:

- **Physical characteristics of interfaces and medium:** The physical layer defines the characteristics of the interface between the devices and the transmission medium. It also defines the type of transmission medium.
- **Representation of bits:** The physical layer data consists of a stream of bits (sequence of 0s or 1s) with no interpretation. To be transmitted, bits must be encoded into signals-electrical or optical. The physical layer defines the type of encoding (how 0s and 1s are changed to signals).
- **Data rate:** The transmission rate-the number of bits sent each second-is also defined by the physical layer. In other words, the physical layer defines the duration of a bit, which is how long it lasts.

Physical Layer

- **Synchronization of bits:** The sender and receiver not only must use the same bit rate but also must be synchronized at the bit level. In other words, the sender and the receiver clocks must be synchronized.
- **Line configuration:** The physical layer is concerned with the connection of devices to the media; point-to-point configuration or multipoint configuration.
- **Physical topology:** The physical topology defines how devices are connected to make a network. Devices can be connected by using a mesh topology, a star topology, a ring topology, a bus topology.
- **Transmission mode:** The physical layer also defines the direction of transmission between two devices: simplex, half-duplex, or full-duplex.

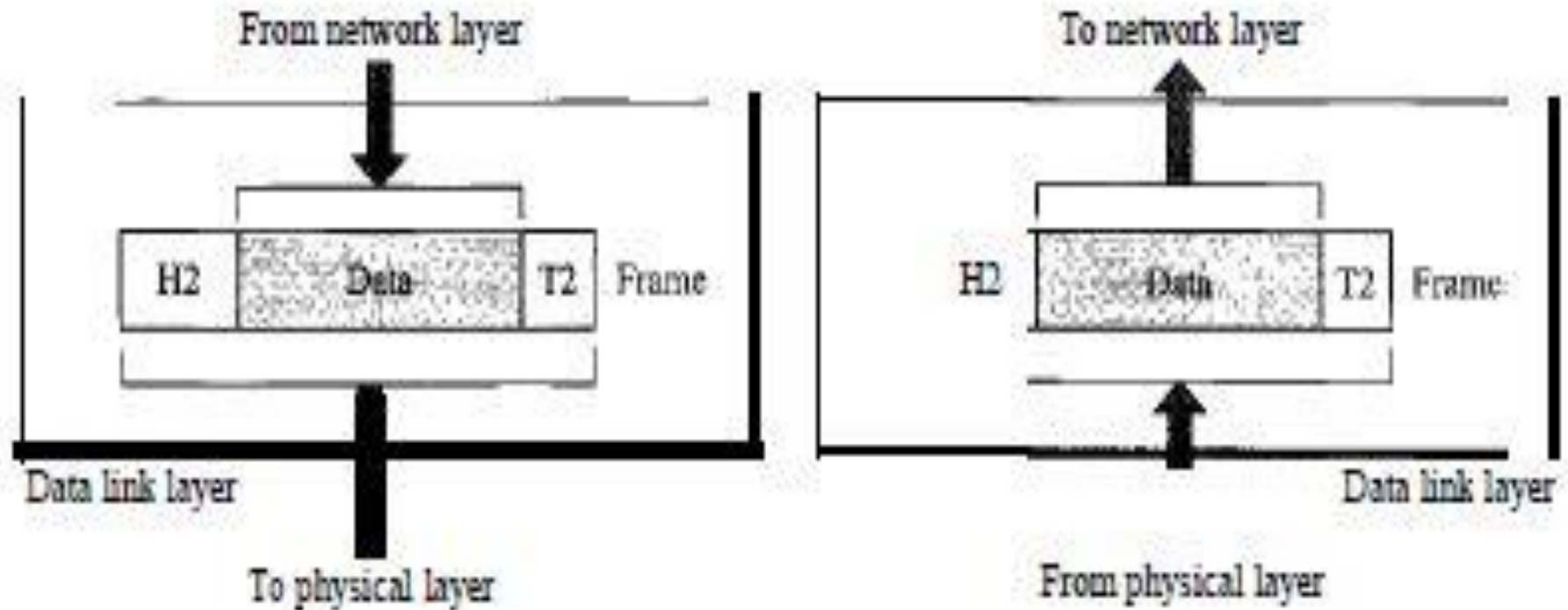
Data Link Layer

➤ Data Link Layer

- Data link layer performs the most reliable node to node delivery of data.
- It forms frames from the packets that are received from network layer and gives it to physical layer.
- It also synchronizes the information which is to be transmitted over the data.
- Error controlling is easily done.
- The encoded data are then passed to physical.

Data Link Layer

Figure 2.6 *Data link layer*



Data Link Layer

Responsibilities of the data link layer include the following:

- **Framing:** The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- **Physical addressing:** If frames are to be distributed to different systems on the network, the data link layer adds a header to the frame to define the sender and/or receiver of the frame. If the frame is intended for a system outside the sender's network, the receiver address is the address of the device that connects the network to the next one.

Data Link Layer

- **Flow control:** If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- **Error control:** The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error control is normally achieved through a trailer added to the end of the frame.
- **Access control.** When two or more devices are connected to the same link, data link layer protocols are necessary to determine which device has control over the link at any given time.

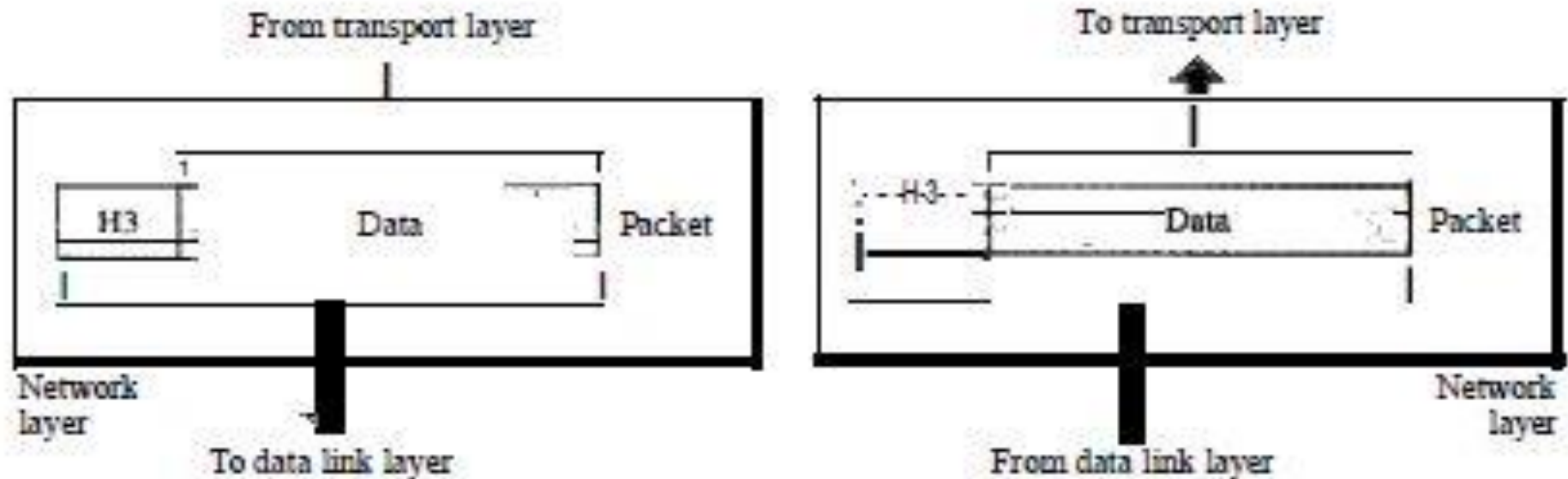
Network Layer

➤ Network Layer

- The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links).
- Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.
- If two systems are connected to the same link, there is usually no need for a network layer.
- However, if the two systems are attached to different networks (links) with connecting devices between the networks (links), there is often a need for the network layer to accomplish source-to-destination delivery.

Network Layer

Figure 2.8 *Network layer*



The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Network Layer

Responsibilities of the network layer include the following:

- **Logical addressing:** The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.

Network Layer

- **Routing:** When independent networks or links are connected to create internetworks (network of networks) or a large network, the connecting devices (called routers or switches) route or switch the packets to their final destination. One of the functions of the network layer is to provide this mechanism.

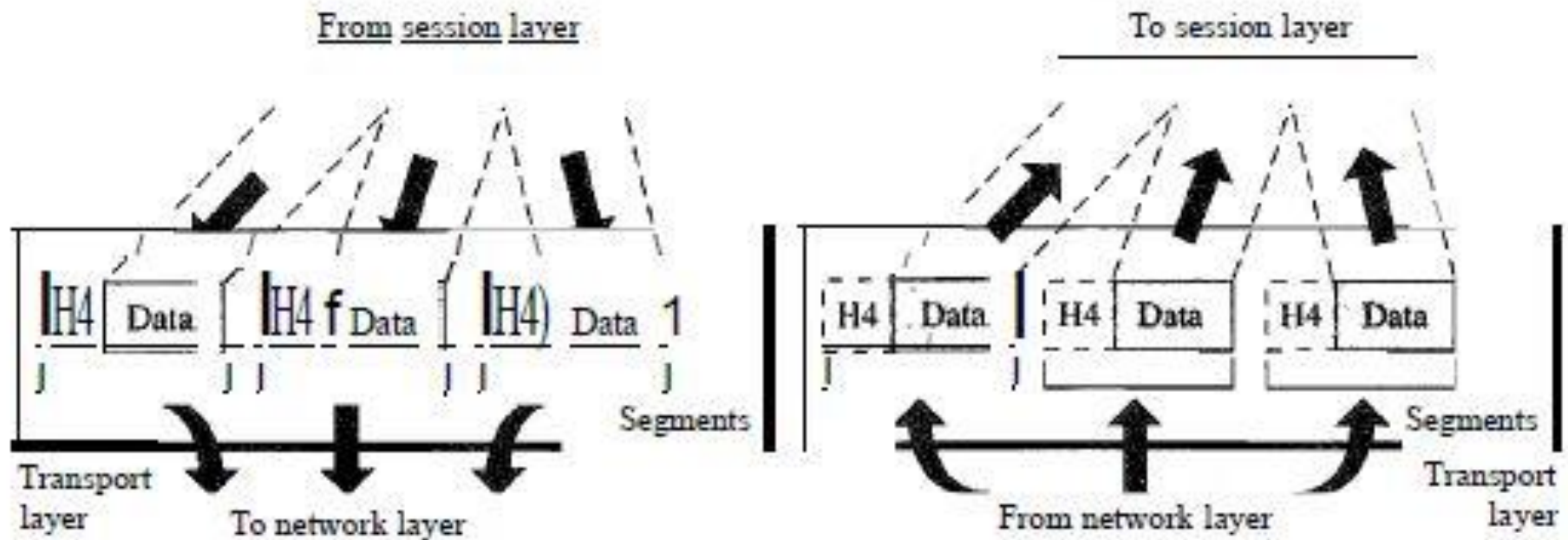
Transport Layer

➤ Transport Layer

- The transport layer is responsible for process-to-process delivery of the entire message.
- A process is an application program running on a host.
- Whereas the network layer oversees source-to-destination delivery of individual packets, it does not recognize any relationship between those packets.
- It treats each one independently, as though each piece belonged to a separate message, whether or not it does.
- The transport layer, on the other hand, ensures that the whole message arrives intact and in order, overseeing both error control and flow control at the source-to-destination level.

Transport Layer

Figure 2.10 *Transport layer*



The transport layer is responsible for the delivery of a message from one process to another.

Transport Layer

Responsibilities of the transport layer include the following:

- **Service-point addressing:** Computers often run several programs at the same time. For this reason, source-to-destination delivery means delivery not only from one computer to the next but also from a specific process (running program) on one computer to a specific process (running program) on the other. The transport layer header must therefore include a type of address called a service-point address (or port address). The network layer gets each packet to the correct computer; the transport layer gets the entire message to the correct process on that computer.
- **Segmentation and reassembly:** A message is divided into transmittable segments, with each segment containing a sequence number. These numbers enable the transport layer to reassemble the message correctly upon arriving at the destination and to identify and replace packets that were lost in transmission.

Transport Layer

- **Connection control:** The transport layer can be either connectionless or connection-oriented. A connectionless transport layer treats each segment as an independent packet and delivers it to the transport layer at the destination machine. A connection-oriented transport layer makes a connection with the transport layer at the destination machine first before delivering the packets. After all the data are transferred, the connection is terminated.
- **Flow control:** Like the data link layer, the transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.
- **Error control:** Like the data link layer, the transport layer is responsible for error control. However, error control at this layer is performed process-to-process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

Protocols of Transport Layer

- This link between the application and transport layer protocols is based on the ports selected to support the individual sessions supported by the transport layer.
- There are 65,535 port numbers available, and these are divided into three ranges: Well-known ports (0 to 1023). Dynamic or private ports (49152 to 65535).
- Port numbers 1 to 1023 are well known and assigned by the Internet Assigned Numbers Authority (IANA) to standardized services running with root privileges.
- For example, Web servers listen on TCP port 80 for client requests. Port numbers 1024 to 49151 are used by various applications with ordinary user privileges.

Number	Assignment
119	Network News Transfer Protocol (NNTP)
123	Network Time Protocol (NTP)

Transport Layer

Some important port addresses:

- FTP (TCP Port 21)
- SSH (TCP Port 22)
- Telnet (TCP port 23)
- SMTP email (TCP port 25)
- HTTP (TCP port 80)
- HTTPS (TCP port 443)
- POP3 email (TCP port 110)
- DNS (TCP or UDP port 53)
- TFTP (UDP port 69)
- SNMP (UDP port 161)
- RIP (UDP port 520)

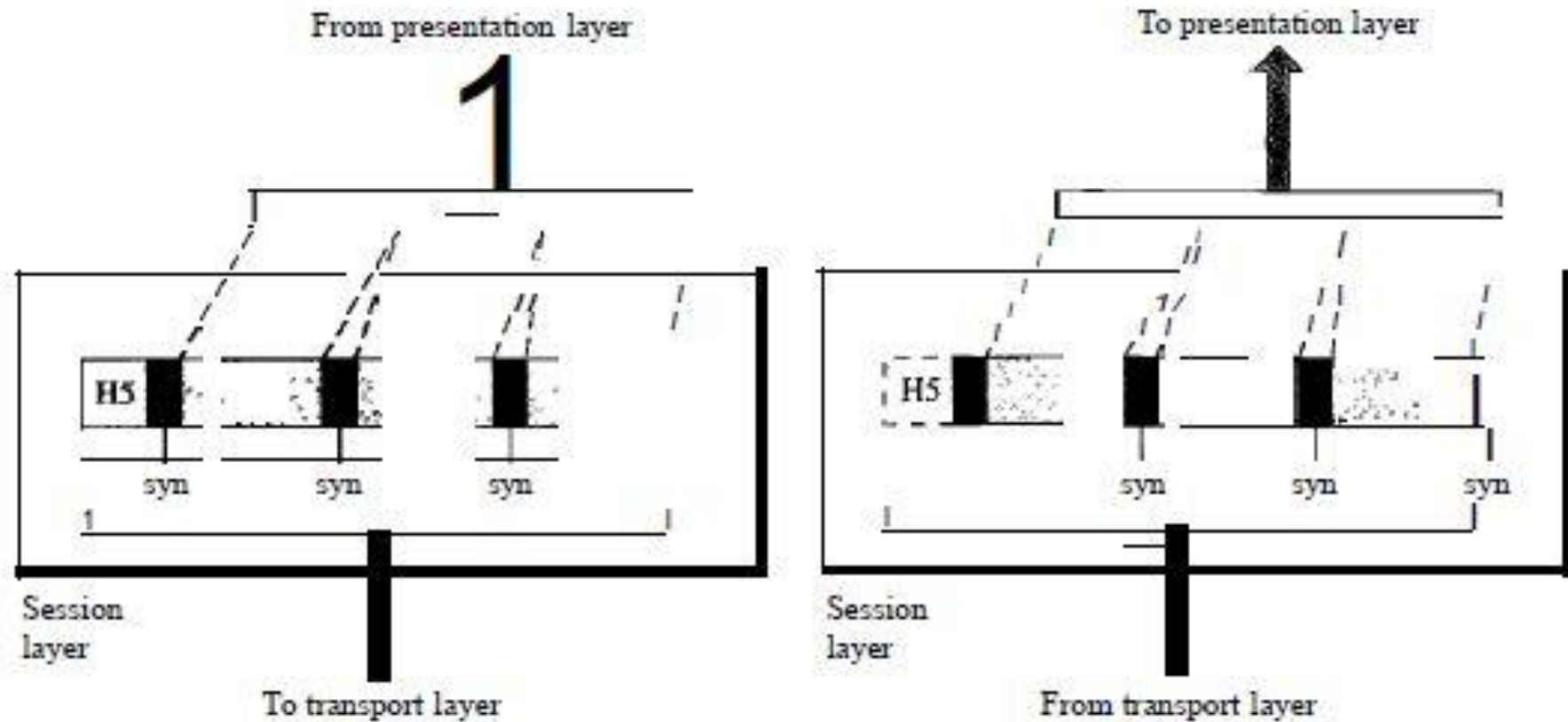
Session Layer

➤ Session Layer

- The session layer provides the mechanism for opening, closing and managing a session between end-user application processes.
- Communication sessions consist of requests and responses that occur between applications.
- Session-layer services are commonly used in application environments that make use of remote procedure calls (RPCs).

Session Layer

Figure 2.12 *Session layer*



Session Layer

- An example of a session-layer protocol is the OSI protocol suite session-layer protocol, also known as X.225 or ISO 8327.
- In case of a connection loss this protocol may try to recover the connection.
- If a connection is not used for a long period, the session-layer protocol may close it and re-open it.
- It provides for either full duplex or half-duplex operation and provides synchronization points in the stream of exchanged messages.

Session Layer

- Other examples of session layer implementations include Zone Information Protocol (ZIP) – the AppleTalk protocol that coordinates the name binding process, and Session Control Protocol (SCP) – the DECnet Phase IV session-layer protocol.

Session Layer

Responsibilities of the session layer include the following:

- **Dialog control:** The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half-duplex or full-duplex .
- **Synchronization:** The session layer allows a process to add checkpoints, or synchronization points, to a stream of data. For example, if a system is sending a file of 2000 pages, it is advisable to insert checkpoints after every 100 pages to ensure that each 100-page unit is received and acknowledged independently. In this case, if a crash happens during the transmission of page 523, the only pages that need to be resent after system recovery are pages 501 to 523. Pages previous to 501 need not be resent. Figure 2.12 illustrates the relationship of the session layer to the transport and presentation layers.

Presentation Layer

➤ Presentation Layer

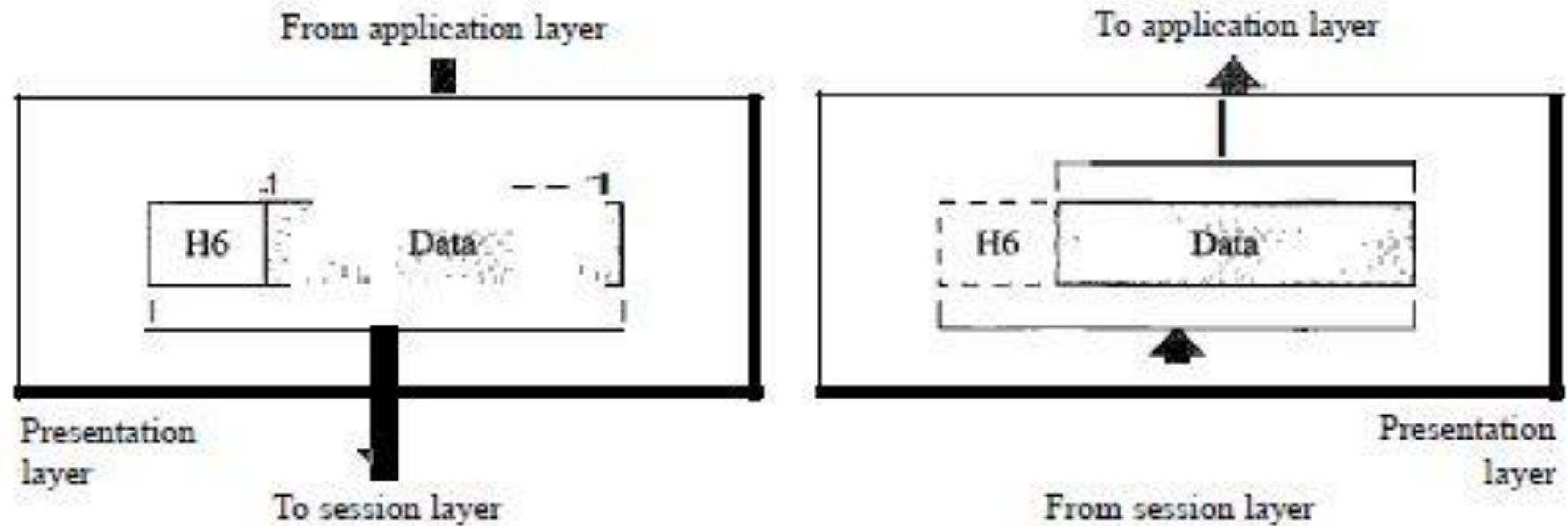
- The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.

Responsibilities of the presentation layer include the following:

- **Translation:** The processes (running programs) in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted. Because different computers use different encoding systems, the presentation layer is responsible for interoperability between these different encoding methods. The presentation layer at the sender changes the information from its sender-dependent format into a common format. The presentation layer at the receiving machine changes the common format into its receiver-dependent format.

Presentation Layer

Figure 2.13 *Presentation layer*



The presentation layer is responsible for translation, compression, and encryption.

Presentation Layer

- **Encryption:** To carry sensitive information, a system must be able to ensure privacy. Encryption means that the sender transforms the original information to another form and sends the resulting message out over the network. Decryption reverses the original process to transform the message back to its original form.
- **Compression:** Data compression reduces the number of bits contained in the information. Data compression becomes particularly important in the transmission of multimedia such as text, audio, and video.

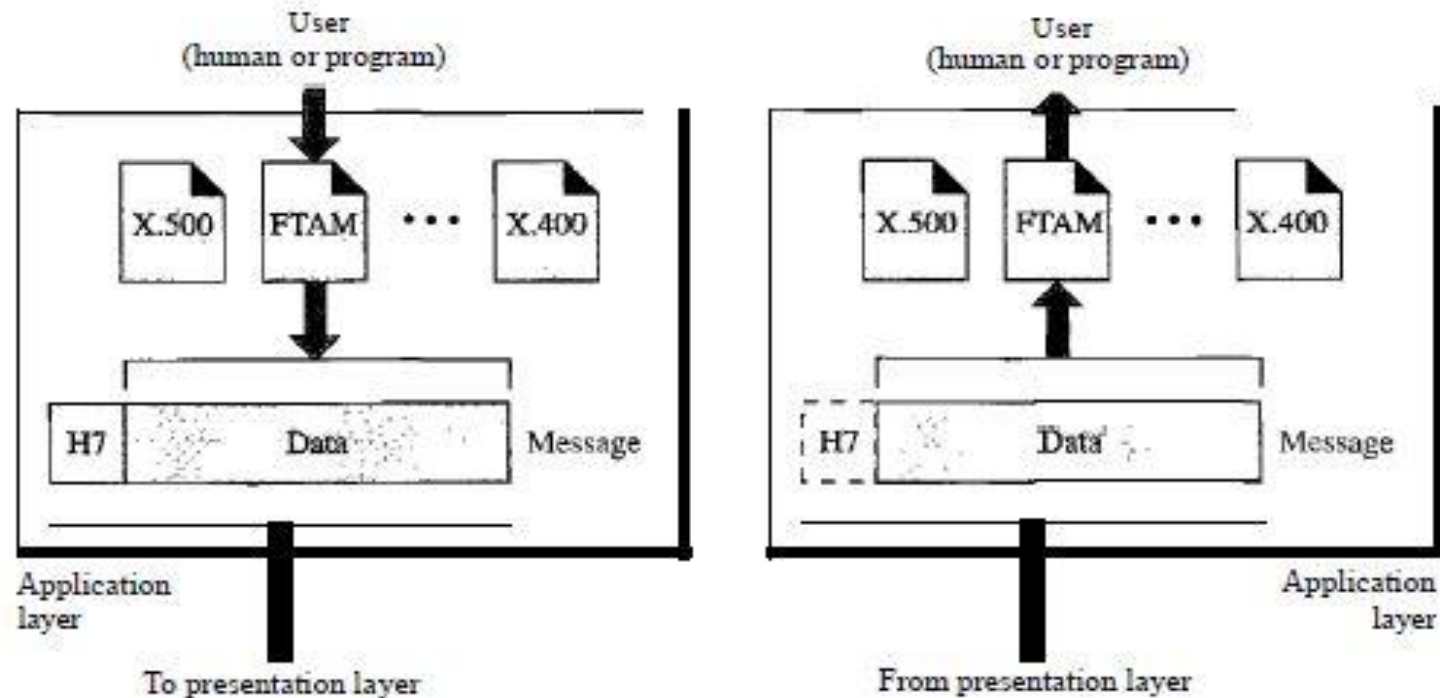
Application Layer

➤ Application Layer:

- The application layer enables the user, whether human or software, to access the network.
- It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- Figure 2.14 shows the relationship of the application layer to the user and the presentation layer.
- Of the many application services available, the figure shows only three: X.400 (message-handling services), X.500 (directory services), and file transfer, access, and management (FTAM). The user in this example employs X.400 to send an e-mail message.

Application Layer

Figure 2.14 *Application layer*



The application layer is responsible for providing services to the user.

Application Layer

Specific services provided by the application layer include the following:

- **Network virtual terminal:** A network virtual terminal is a software version of a physical terminal, and it allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host. The user's computer talks to the software terminal which, in turn, talks to the host, and vice versa. The remote host believes it is communicating with one of its own terminals and allows the user to log on.

Application Layer

- **File transfer, access, and management:** This application allows a user to access files in a remote host (to make changes or read data), to retrieve files from a remote computer for use in the local computer, and to manage or control files in a remote computer locally.
- **Mail services:** This application provides the basis for e-mail forwarding and storage.
- **Directory services:** This application provides distributed database sources and access for global information about various objects and services.

Protocols of Application Layer

Some protocols of Application Layer are given:

TELNET	TELeType NETwork (23)	POP3	Post Office Protocol 3
DNS	Domain Name System (53)	BOOTP	Bootstrap Protocol
DHCP	Dynamic Host Configuration Protocol (67, 68)	XMPP	Extensible Messaging and Presence Protocol
FTP	File Transfer Protocol (20 for data and 21 for control)	TFTP	Trivial File Transfer Protocol (69)
SMTP	Simple Mail Transfer Protocol (25)	LPD	Line Printer Daemon (515)
HTTP	Hyper Text Transfer Protocol	CoAP	Constrained Application Protocol
NFS	network file system (2049)	DDS	Data Distribution Service
X window	a protocol for the writing of graphical user interface–based client/server applications (Port number for X window starts from 6000 and increases by 1 for each server).		

Protocols of Application Layer

Some protocols of Application Layer are given:

MQTT	Message Queuing Telemetry Transport	SOAP	Simple Object Access Protocol
SMQTT	Secure Message Queue Telemetry Transport	XMPP-IOT	XMPP Internet of Things
AMQP	Advanced Message Queuing Protocol	LWM2M	Lightweight Machine-to-Machine
MQTT-SN	MQTT for Sensor networks	SSI	Simple Sensor Interface
SMCP	Standard Marine Communication Phrases	RESTful HTTP	REpresentational State Transfer
LLAP	Lightweight Logical Automation Protocol	CMOT	Common Management Information Protocol
SNMP	Simple Network Management Protocol (161 (TCP) and 162 (UDP))		

Protocols of Application Layer

Some protocols of Application Layer are given:

HTTPS	Hypertext Transfer Protocol Secure	DeviceNet	Automation Industry Protocol
NTP	Network Time Protocol	eDonkey	Classic File Sharing Protocol
SSH	Secure Shell Protocol	H.323	Packet-Based Multimedia Communications System
IMAP	Internet Message Access Protocol	LDAP	Lightweight Directory Access Protocol
IRC	Internet Relay Chat	NIS	Network Information Service
VTP	Virtual Terminal Protocol	RDP	Remote Desktop Protocol
SSMS	Secure SMS Messaging Protocol	RTP	Real-time Transport Protocol
SMP	Server Message Block	SIP	Session Initiation Protocol
MIME(S-MIME)	Multipurpose Internet Mail Extensions and Secure MIME		

Protocols of Presentation Layer

Some protocols of Presentation Layer are given:

MPEG		ASCH	
SSL	Secure Socket Tunneling	TLS	Transport Layer Security
AFP	Apple Filing Protocol	ICA	Independent Computing Architecture
LPP	Lightweight Presentation Protocol	NCP	NetWare Core Protocol
NDR	Network Data Representation	XDR	eXternal Data Representation
PAD	X.25 Packet Assembler/Disassembler Protocol		

Protocols of Session Layer

Some protocols of Session Layer are given:

NetBIOS	File Sharing and Name Resolution protocol	SAP	Session Announcement Protocol
ASP	AppleTalk Session Protocol	NCP	NetWare Core Protocol
ADSP	AppleTalk Data Stream Protocol	PAP	Printer Access Protocol
NetBEUI	NetBIOS Enhanced User Interface	RPC	Remote Procedure Call
SMB	Server Message Block	RTCP	RTP Control Protocol
SMPP	Short Message Peer-to-Peer	SDP	Sockets Direct Protocol
ZIP	Zone Information Protocol	iSNS	Internet Storage Name Service
SOCKS	SOCKS is an Internet protocol that exchanges network packets between a client and server through a proxy server.		

Protocols of Transport Layer

■ Popular Transport Layer Protocols

ATP	AppleTalk Transaction Protocol	SCTP	Stream Control Transmission Protocol
CUDP	Cyclic UDP	SPX	Sequenced Packet Exchange
DCCP	Datagram Congestion Control Protocol	SST	Structured Stream Transport
FCP	Datagram Congestion Control Protocol	TCP	Transmission Control Protocol
IL	Fibre Channel Protocol	UDP	User Datagram Protocol
MPTC P	IL Protocol	UDP-Lite	User Datagram Protocol
RDP	Multipath TCP	μTP	Micro Transport Protocol
RUDP	Reliable User Datagram Protocol		

Protocols of Transport Layer

■ Popular Transport Layer Protocols

AEP	AppleTalk Echo Protocol	FCP	Fibre Channel Protocol
DCCP	Datagram Congestion Control Protocol	ESP	Encapsulating Security Payload
iSCSI	Internet Small Computer System Interface	SCTP	Stream Control Transmission Protocol
NBF	NetBIOS Frames protocol	TUP	Telephone User Part
SPX	Sequenced Packet Exchange	NBP	Name Binding Protocol

TCP Vs. UDP

Basis for Comparison	TCP	UDP
Definition	TCP establishes a virtual circuit before transmitting the data.	UDP transmits the data directly to the destination computer without verifying whether the receiver is ready to receive or not.
Connection Type	It is a Connection-Oriented protocol	It is a Connectionless protocol
Speed	slow	high
Reliability	It is a reliable protocol.	It is an unreliable protocol.
Header size	20 bytes	8 bytes
acknowledgement	It waits for the acknowledgement of data and has the ability to resend the lost packets.	It neither takes the acknowledgement, nor it retransmits the damaged frame.

Protocols of Network Layer

Some protocols of Network Layer are given:

IP	Internet protocol	ARP	Address Resolution Protocol
ICMP	Internet Control Message Protocol	IPSEC	Internet Protocol Security
MPLS	Multiprotocol Label Switching	IGMP	Internet Group Management Protocol
EGP	Exterior Gateway Protocol	DDP	Datagram Delivery Protocol
EIGRP	Enhanced Interior Gateway Routing Protocol	LLARP	Low Latency Anonymous Routing Protocol
IPX	Internetwork Packet Exchange	NAT	Network Address Translation
OSPF	Open Shortest Path First	PIM	Protocol Independent Multicast
RIP	Routing Information Protocol	SCCP	Signalling Connection Control Part
TIPC	Transparent Inter-process Communication	PPPoE	Point-to-Point Protocol over Ethernet
HSRP	Hot Standby Router protocol	VRRP	Virtual Router Redundancy Protocol

Protocols of Network Layer

Some protocols of Network Layer are given:

ATM:- Asynchronous Transfer Mode.

IS-IS:- Intermediate System - Intermediate System (OSI).

SPB:- Shortest Path Bridging.

MTP:- Message Transfer Part.

NSP:- Network Service Part.

TRILL:- TRansparent Interconnection of Lots of Links.

Protocols of Data Link Layer

Some protocols of Data Link Layer are given:

PAP	Password Authentication Protocol	SDLC	Synchronous Data Link Protocol
PPTP	Point-to-Point Tunneling Protocol	HDLC	High Level Data Link Protocol
ARP	Address Resolution Protocol	SLIP	Serial Line Interface Protocol
LLDP	Link Layer Discovery Protocol	PPP	Point to Point Protocol
FDDI	Fiber Distributed Data Interface	LCP	Link Control Protocol
LAP	Link Access Procedure	NCP	Network Control Protocol
NDP	Neighbour Discovery Protocol	STP	Spanning Tree Protocol
CDP	Cisco Discovery Protocol	VTP	VLAN Trunking Protocol
MPLS	Multiprotocol Label Switching	UDLD	Unidirectional Link Detection

Protocols of Physical Layer

Some protocols of Physical Layer are given:

RS232, 100BaseTX, ISDN, 11.

USB	Universal Serial Bus	SPI	Serial Peripheral Interface
SONET	Synchronous Optical Networking	SDH	Synchronous Digital Hierarchy
SMB	System Management Bus	ISDN	Integrated Services Digital Network
100BASE-T	Fast Ethernet	ADVB	Avionics Digital Video Bus
CAN	Controller Area Network.	DSL	Digital subscriber line
Bluetooth		Fiber cables	

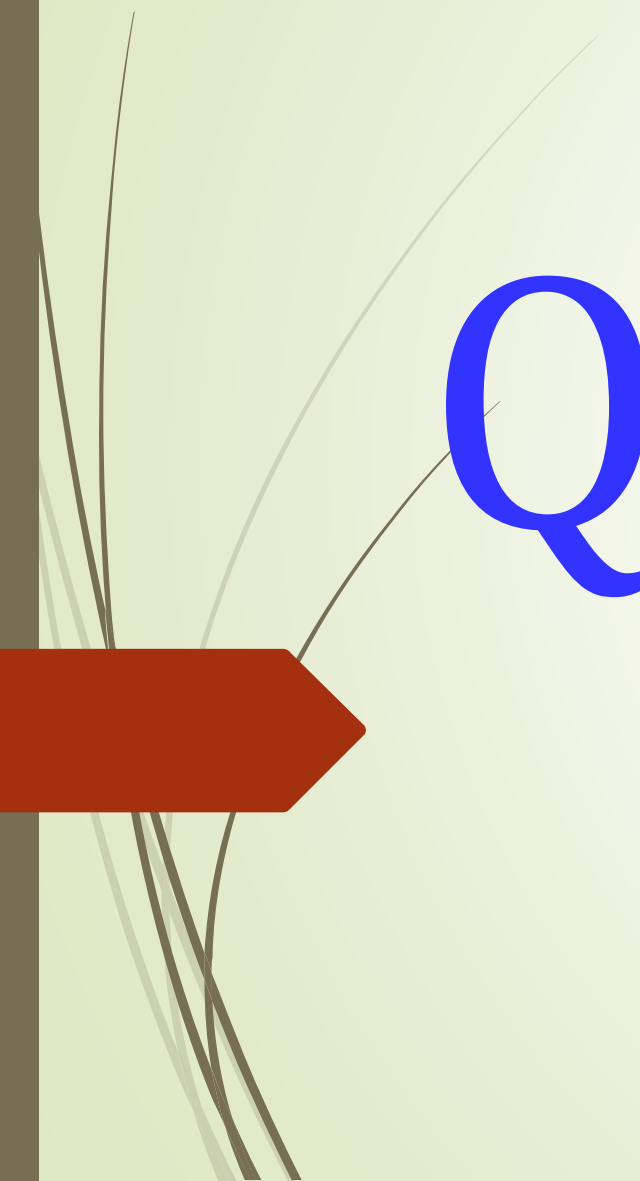
OSI Vs. TCP/IP model

- Here, are some important differences between the OSI & TCP/IP model:

OSI Model	TCP/IP model
OSI model provides a clear distinction between interfaces, services, and protocols.	TCP/IP doesn't offer any clear distinguishing points between services, interfaces, and protocols.
OSI uses the network layer to define routing standards and protocols.	TCP/IP uses only the Internet layer.

OSI Vs. TCP/IP model

OSI Model	TCP/IP model
OSI model use two separate layers physical and data link to define the functionality of the bottom layers	TCP/IP uses only one layer (link).
OSI model, the transport layer is only connection-oriented.	A layer of the TCP/IP model is both connection-oriented and connectionless.
In OSI model, data link layer and physical are separate layers.	In TCP data link layer and physical layer are combined as a single host-to-network layer.
The minimum size of the OSI header is 5 bytes.	Minimum header size is 20 bytes.



Questions?



Thanks All